

The Next LEVEL

	2G	3G
Voice Calling	Yes	Yes, with improved voice quality
Video Calling	No	Yes, video conferencing as well
Network	Saturated, regular call drops	Congestion-free network, less call drops
Data Transfer	GPRS: 56 Kbps up to 115 Kbps EDGE: Up to 236.8 Kbps	2 Mbps to 7 Mbps
Gaming	Single player on phone	Multiplayer interactive gaming, online gaming
Television	Slow streaming rates	Fast buffer speed, real time broadcast
Music Download	Very slow download, when possible	Fast downloads
Video Download	Files are compressed resulting in impaired video quality	High speed downloads and high quality, hi-definition video
Location Based Services	Download maps, search locations	Routing, real-time traffic updates, location-based gaming, social networking
e-commerce	Transactions over USSD*	Net banking, stock trading
Value Added Services	Ring back tones, astrology, voice-based services	Content on demand, graphically enhanced wide services
Rural Services	Limited services via SMS	Telemedicine, agriculture applications, weather updates, mobile education, community-based enhancements
Price**	Voice calls: 30 paise/min; data: Rs 95/month onwards	Video calling: 30 paise to Re 1/min; data: 1 paisa/10 Kbps

*Unstructured Service Supplementary Data: **Current MTNL prices

Stock trading will also form a domain on the phone. Real-time information can be accessed and transaction carried out on the move.

► The download speeds can go up to 7 Mbps (Megabits per second). This translates into a download time of seven seconds for a 3 MB MP3 song. Compare that to eight minutes required

at 2G speeds. Music videos will become a staple on mobile phones.

► Auctions are also being held for broadband wireless access (BWA) that will usher in a faster-than-3G WiMAX and LTE. (The West is already moving to 4G, which offers speeds up to 10 times faster than 3G).

by Nidhi Singal

might help in bringing down the deficit by a couple of basis points. Earlier, the reserve price for a 3G pan-India spectrum licence was Rs 2,020 crore, which was raised last year to Rs 3,500 crore."

One of the biggest fallouts of telecom operators winning 3G spectrum will be the decongestion of the 2G spectrum. So expect fewer call drops. Says Gaurav Tyagi, analyst at Alchemy Capital Management: "In the initial months, operators will use 3G spectrum to decongest the 2G space. Consumers contributing high average revenues will be migrated to the 3G space for a fee. This will free up space in the 2G network."

That may be good news for users, but it may not necessarily be glad tidings for telecom's big boys. None of them bid for all 22 circles—clearly, the pan-India price of 3G did not look viable to any operator. Even the big winners—Bharti, Reliance and Vodafone—chose to omit a few circles. Bharti will pay \$2.8 billion (Rs 32 per share) for 13 circles, Reliance \$1.9 billion (Rs 40 per share) for 13 circles and Idea will pay \$1.3 billion (Rs 18 per share) for 11 circles.

But is there enough upside from 3G for these companies to justify the price they have paid? Analysing the impact of 3G on industry and telcos, a Kotak Securities report says, "The auctions for three slots in 17 circles and four slots in the other five circles of 2,100 MHz of 3G spectrum have come to a close with the pan-India clearing price settling at Rs 16,700 crore (\$3.7 billion). The overall bid price surpassed our 'assuming rationality' expectation by \$1.2 billion." So why did the telecom players pay an amount so much higher than expected? Industry experts believe that fear drove the contest. While the prize of 3G may not be worth the price, the possible punishment (fall in stock prices at not winning 3G spectrum) made the fight seem worth it. Operators have based their auction decision on the price they would pay by losing high ARPU (Average Revenue per User) customers rather than the potential value accretion from owning 3G. Jaideep Ghosh, executive director, KPMG, explains: "The bids for